PATTERN RECOGNITION
CASE STUDY / MINI PROJECT REPORT

15 Marks | Two Reviews – 7.5 Marks Each

Student and Project Details

Course Code / Course Title	21CSE451T – Pattern Recognition Techniques
Academic Year / Semester	2026–27 (Odd) / Semester 7
Department / Section	Computational Intelligence (CINTEL) / AC-1
Student Name(s)	Vansh Agrawal
Register Number(s)	RA2311026010120
Project Title	AI-Based Archaeological Pottery Fragment Matching
Faculty Guide / Reviewer	Dr Nithyashri J
Review Date(s)	Review 1: _____ Review 2: _____

Project Report

Provide concise, evidence-based explanations. Students must explain their own understanding and justify the choices made in the case study. Attach relevant figures, tables, graphs, dataset samples and references where appropriate.

1. Abstract

Summarise the problem, dataset, pattern recognition approach, key results and main conclusion (approximately 150–250 words).

Response:

Archaeological pottery is usually excavated as hundreds of broken sherds, and finding which sherds physically join is slow expert work. This case study treats joining as a pattern-recognition problem: given one sherd, rank every other sherd in the collection by the probability that the two pieces fit together. We use the Daxinzhuang-18K benchmark (Ji et al., 2026). Its gold standard of 124 expert-photographed sherds with 111 verified joins (from 33 vessel groups) is the test set. We also use 2,212 simulated fracture pairs and the excavation metadata. Each sherd is segmented, illumination-normalised and described by colour, LBP texture, ResNet-50 and DINOv2 embeddings. Each pair of sherds is also checked for fracture-edge fit: complementary curvature windows are aligned rigidly (Kabsch with ICP refinement) and scored by edge contact, area overlap and surface-stripe orientation. A class-balanced logistic regression fuses the twelve pair features into a join probability. Under leak-free 5-fold vessel-grouped cross-validation, the proposed model retrieves a true partner in the top 5 for 55.6% of queries and in the top 10 for 71.8%. The image-only DINOv2 baseline from Review 1 reaches 40.3% and 54.0%, a significant improvement of +17.7 pts (95% CI +9.7 to +26.6) at top 10. However, most of the gain comes from learning to combine appearance cues. Adding geometry changes Hit@5 by only +3.2 pts (95% CI -2.4 to +9.7), and it lowers Hit@1. The same edge matcher recovers the correct pose for 47.0% of clean synthetic breaks but only 17.1% of real ones. We conclude that a learned appearance ranker is a practical triage tool. Fracture geometry needs edge-wear-robust descriptors before it can reliably help on eroded real sherds.

Keywords: pottery sherd matching, fragment reassembly, image retrieval, contour matching, logistic regression, cultural heritage.

2. Introduction and Problem / Case Study Understanding

Describe the real-world problem or application. State the problem clearly, identify the pattern to be recognised/classified/detected, explain why it matters and define the project objectives.

Response:

2.1 Real-world problem

Pottery is the most common find on most excavations. At the Bronze-Age site of Daxinzhuang (Jinan, Shandong, China), the excavation catalogue alone lists 17,635 sherds from one refuse pit. Before a vessel can be studied for typology, capacity, use-wear or dating, its sherds must be re-associated. Today an archaeologist spreads the sherds on a table and compares them one against another. The work scales with the square of the number of pieces, so it is the first thing to be cut when time or funding runs short. Many collections therefore stay as unsorted bags of fragments.

2.2 The pattern to be recognised

The pattern is a **relation between two objects**, not a property of one object. A pair of sherds either *joins* (the pieces were neighbours on the original vessel and share a fracture edge) or does not. We formulate it in two equivalent ways:

- **Pair classification**: input = two sherd images; output = P(join). This is a binary pattern-recognition problem with extreme class imbalance: in the gold standard only 111 of 7,626 possible pairs (1.46%) join.
- **Retrieval / ranking**: input = one query sherd; output = all other sherds sorted by P(join). This is how the system is used: an archaeologist inspects the top-k candidates instead of the whole collection.

2.3 Why it is hard

- **Look-alike sherds.** Most Daxinzhuang sherds are the same grey-to-orange cord-marked ware, so sherds from different vessels look alike (Figure 1).
- **Eroded edges.** Fracture edges are chipped and worn after ~3,000 years in the ground, so even true partners leave visible gaps when assembled (Figure 6).
- **No canonical orientation or position.** A sherd can be photographed at any rotation, so every comparison must be rotation-invariant or must search over rotations.
- **Two cues that can disagree.** Two pieces may look alike but not fit (same ware, different vessel), or fit geometrically by coincidence (straight edges fit many places).

2.4 Research question carried over from Review 1

Does adding fracture-edge geometry to image similarity improve top-k matching over image-only retrieval? Hypothesis H1: a model that combines appearance and fracture-edge geometry retrieves true partners at higher Hit@k than appearance alone, because geometry can reject look-alike sherds that cannot physically fit.

2.5 Related work

Fragment reassembly is one of the oldest problems in pattern recognition. Freeman and Garder (1964) solved apictorial jigsaw puzzles by matching boundary shape alone, and Wolfson (1990) formalised curve matching for such boundaries. In 3-D, Papaioannou et al. (2002) and Huang et al. (2006) reassembled fractured objects by matching fracture surfaces. Those methods depend on clean, high-resolution fracture geometry. For pottery specifically, Kampel and Sablatnig (2003) used profile sections to reconstruct vessel shape. Recent 2-D learning work mostly *classifies* sherds rather than *joining* them. Chetouani et al. (2020) combined pre-trained CNN features to classify engraved sherd patterns. Pinho et al. (2025) used a pre-trained ResNet-50 to group sherds by surface pattern and colour to assist reconstruction. Yu et al. (2026) showed that sherd recognisers latch onto spurious background features and used contrastive learning to suppress them. The Daxinzhuang-18K benchmark (Ji et al., 2026) is the first large public 2-D collection with expert-verified physical joins, which makes the comparison in this study possible.

Table 1. Positioning against prior work.

Work	Cue used	Data	What it leaves open for this study
Freeman & Garder 1964; Wolfson 1990	Boundary shape	Synthetic jigsaw pieces	Clean cut edges; no appearance
Papaioannou et al. 2002; Huang et al. 2006	3-D fracture surface	3-D scans	Needs 3-D scanning of every sherd
Kampel & Sablatnig 2003	Profile section	Rim sherds	Body sherds have no usable profile
Chetouani et al. 2020	CNN appearance	Engraved sherds	Classifies patterns, does not find joins
Pinho et al. 2025	ResNet-50 appearance	Decorated sherds	Groups by look; geometry not tested
Yu et al. 2026	Contrastive CNN	Hemudu sherds	Recognition, not reassembly

Work	Cue used	Data	What it leaves open for this study
This study	Appearance + 2-D fracture edge	Daxinzhuang gold standard (real joins)	Tests whether 2-D edge geometry adds value on real eroded sherds

Research gap. Appearance-based sherd work rarely evaluates on real, expert-confirmed joins, and geometry-based reassembly is mostly evaluated on clean or synthetic breaks. Whether 2-D fracture-edge geometry still helps once edges are eroded, compared fairly against a strong image-only baseline, has not been measured. That comparison is the core of this case study.

2.6 Key terms

Table 2. Archaeological and technical terms.

Term	Meaning in this report
Sherd	A broken fragment of a pottery vessel.
Join / refit	Two sherds that were neighbours on the vessel and share a fracture edge (positive pair).
Vessel group	All sherds connected through joins; a connected component of the join graph.
Exterior / interior view	Photograph of the outer or inner surface of a sherd.
Cord marks	Parallel impressions made by a cord-wrapped paddle while shaping the pot; they form stripes.
Spalling	Flaking off of the surface layer, which changes a sherd's appearance.
Query / gallery	The sherd we search partners for / the collection searched.
Hit@k	Share of queries with at least one true partner among the top k suggestions.
MRR	Mean reciprocal rank of the first true partner.
Pose	Rotation and translation that place a sherd in the assembled position.
Kabsch / ICP	Closed-form best rotation between two point sets / its iterative refinement with re-matched points.

3. Objectives

List the specific, measurable objectives of the case study or mini project.

Response:

Table 3. Specific, measurable objectives.

#	Objective	Measurable target / evidence	Status
O1	Acquire and understand the Daxinzhuang-18K data and metadata	Dataset, metadata and annotation formats documented; pose convention verified	Done (Sec. 4–5)
O2	Build a reproducible preprocessing pipeline	Segment, normalise and extract contours for 100% of gold and simulated images	Done (Sec. 6)
O3	Implement an image-only baseline (Review 1 plan)	Hit@k, MRR of ResNet-50 / DINOv2 cosine ranking	Done (Sec. 12)
O4	Implement fracture-edge geometric compatibility	Pose recovery rate on synthetic and real pairs; pair AUC	Done (Sec. 9, 12)
O5	Fuse appearance and geometry in a pattern-recognition model and test H1	Hit@1/5/10, MRR, mAP, P/R/F1 under vessel-grouped CV, with 95% bootstrap CIs	Done (Sec. 11–13)
O6	Justify the model against alternatives	Logistic regression vs Random Forest vs SVM-RBF, plus an ablation study	Done (Sec. 10)
O7	Deliver a working prototype	Interactive demo that ranks candidates and previews the best physical fit	Done (app.py)

4. Dataset and Data Understanding

Specify the data source, collection method, dataset size, number of samples, features/attributes, classes/labels, data types and important characteristics. Include a sample or summary table if useful.

Response:

4.1 Source and collection

Daxinzhuang-18K: A Large-Scale 2D Benchmark for Pottery Sherd Retrieval and Reassembly, by Ji, Xue, Wang, Ma, Lang, Fang, Tu and Lu, published on Zenodo (record 21553799, version 1.0.0, July 2026, licence CC-BY 4.0). The sherds come from pit H690 at the Daxinzhuang site in Jinan. Each sherd was photographed on a plain backdrop with a colour/scale card, exterior and interior separately, and catalogued by stratigraphic layer. Joins were confirmed physically by experts.

Table 4. Contents of the Zenodo record and how each part is used.

Part	Content	Size	Used here for
Daxinzhuang-18K (real)	36,532 JPG photos (1024×771) + metadata workbook, 17,635 catalogue records	4.7 GB	Metadata study; raw-photo segmentation (E6 planned)
dataset_gold_standard	248 background-removed RGBA PNGs (124 sherds × exterior/interior), gt_all.json with 111 join records and assembly poses	53 MB	Main train/test set (E1–E3, E5)
dataset_simulated (8 subsets)	50,736 synthetic fracture pairs, each with two PNGs and the relative pose	18 GB	sim_data_8 only (2,212 pairs): geometry sanity check (E4)
expert_confirmed_physical_refit_pairs.xlsx	251 joining groups (pairs or triples) by sherd ID	16 KB	Ground truth for the planned raw-photo test (E6)

Because the full record is about 20 GB, we downloaded only what each experiment needs. From the 4.7 GB real-photo archive, the metadata workbook and the needed photos were extracted with HTTP byte-range requests against the zip's central directory, verifying each file's CRC-32. This avoided a full download.

4.2 Samples, labels and features

The unit of analysis is a **sherd pair**. The gold standard has 124 exterior sherds, giving 124·123/2 = 7,626 unordered pairs, of which 111 are labelled *join* (class 1) and the rest *no join* (class 0). The join graph splits into 33 connected components (vessel groups) of 2 to 11 sherds (median 3). Most sherds have one or two partners (56 sherds have exactly one, 44 have two, 18 have three, 6 have four).

Table 5. Gold-standard dataset summary (computed by scripts/explore_dataset.py).

Property	Value
Images (gold)	248 RGBA PNG, 1024 × 1024, transparent background
Sherds used (exterior view)	124
Positive pairs / all pairs	111 / 7,626 (1.46%)
Vessel groups	33 (largest 11 sherds)
Median sherd area	82 k px (full resolution)
Median sherd perimeter	1254 px
Median cord-mark stripe coherence	0.49 (0 = no direction, 1 = perfect stripes)
Simulated pairs (sim_data_8)	2,212 from 1 source vessel, many fracture runs

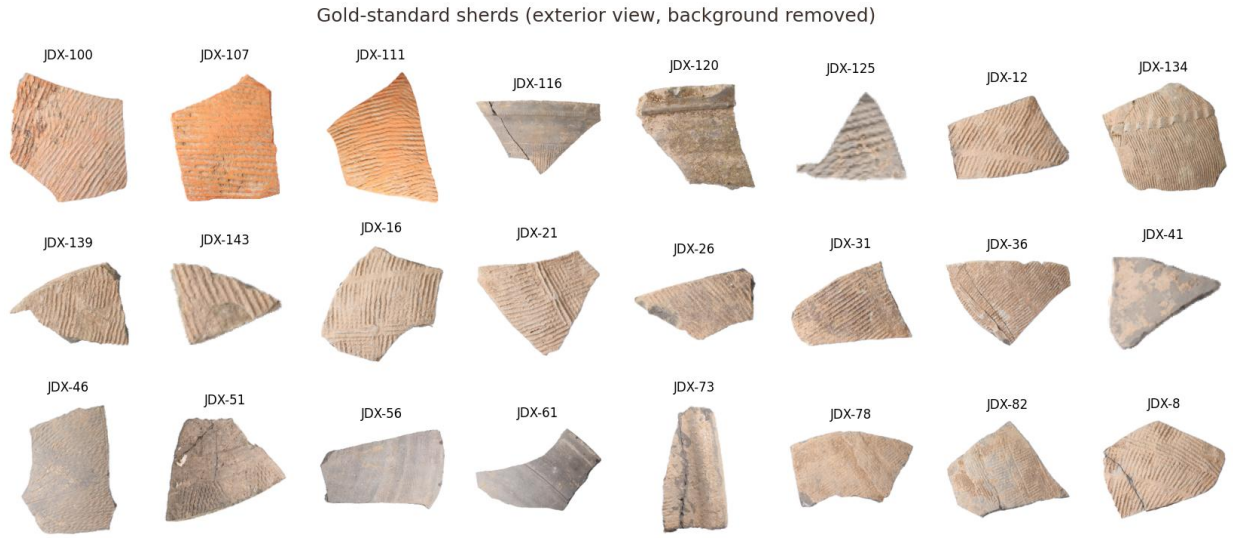


Figure 1. A sample of gold-standard sherds. The ware is uniform: cord-marked, grey to orange, which is why appearance alone confuses vessels.

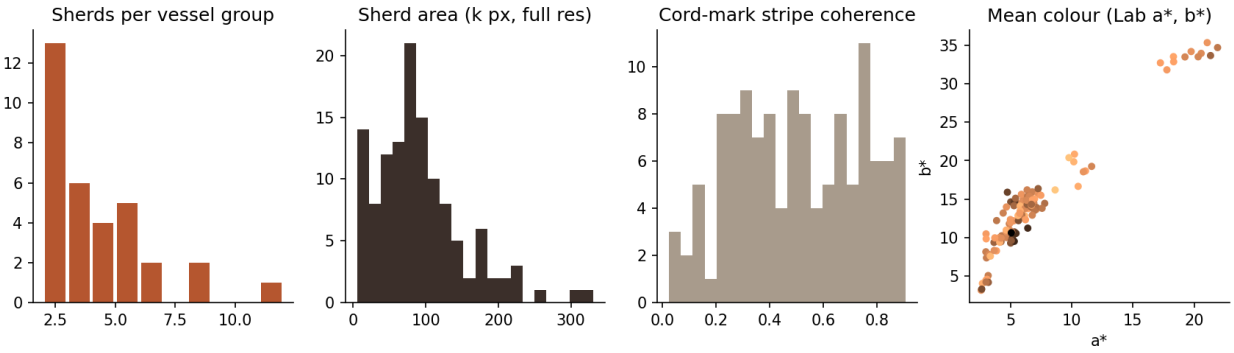


Figure 2. Exploration of the gold standard: vessel-group sizes, sherd areas, strength of the cord-mark stripe pattern, and mean colour in Lab space (colour = lightness).

Observations. (i) Group sizes are skewed: a few large vessels contribute many pairs, so a random pair-level split would put sherds of the same vessel in both train and test. We therefore split by vessel group (Section 11). (ii) Sherd area varies about tenfold, and small sherds have short edges that carry little shape information. (iii) Stripe coherence is moderate (median 0.49), so the stripe direction is a usable but noisy cue. (iv) Colour spreads along a single orange-to-grey axis, the main appearance signal separating vessels.

For each pair we compute 12 features. Four are appearance features and eight describe fracture-edge geometry. All are continuous real numbers:

Table 6. The 12 pair features (inputs to the classifier).

Feature	Type	Meaning
colour_sim	appearance	$-\chi^2$ distance of 8×8×8 Lab colour histograms
lbp_sim	appearance	$-\chi^2$ distance of rotation-invariant uniform LBP histograms (P=8,R=1 and P=16,R=2)
resnet_cos	appearance	cosine similarity of ResNet-50 (ImageNet) 2048-d embeddings
dino_cos	appearance	cosine similarity of DINOv2 ViT-S/14 384-d embeddings
geo_score	geometry	contact × overlap penalty × orientation agreement at the best alignment
contact, contact_frac	geometry	edge length (px) that touches within 4 px, absolute and relative to the shorter outline
overlap	geometry	area overlap of the two sherds after alignment (should be ≈ 0)
rmse	geometry	residual of the touching edge points (px)
orient_gap, orient_conf	geometry	angle between the two sherds' cord-mark directions after alignment, and the confidence of that estimate
log_area_ratio	geometry	|log(area A / area B)|

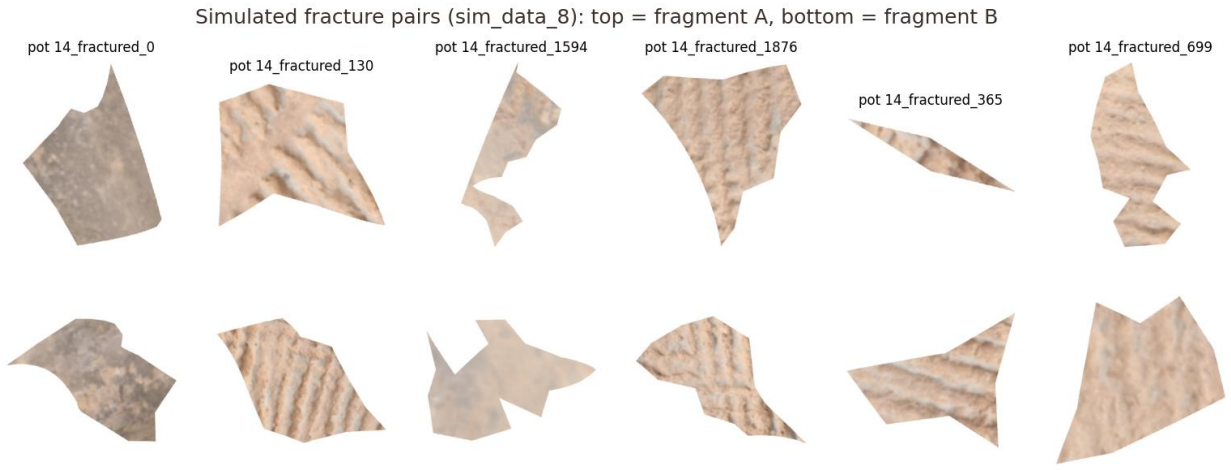


Figure 3. Simulated fracture pairs from sim_data_8. Edges are clean polygon breaks, unlike the eroded real edges.

5. Metadata Understanding

Explain the metadata available (for example: feature names, units, labels, timestamps, source, class descriptions or annotation details) and why it is relevant to interpreting the dataset.

Response:

5.1 Excavation metadata workbook

Daxinzhuang_AI_Archaeological_Sherd_Metadata.xlsx has two sheets: *Shandong* (17,634 records) and *Specimens* (176 records). Row 1 of each sheet is a legend that describes the fields, not a sherd, and must be skipped when the sheet is loaded.

Table 7. Metadata fields and why they matter.

Field	Meaning	Observed values / relevance
Sherd_ID	Catalogue number of the sherd (JD00001…)	Primary key; the refit spreadsheet uses it
Sherd_Part	Rim, body sherd, base, foot, …	Body sherds dominate (15,546); rims 1,501. Rims carry profile information body sherds lack
Inferred_Vessel_Type	Jar, Li tripod, Gui, Dou, …	Only 547 of 17,634 records are typed, so it is too sparse to use as a label
Exterior_Photo / Interior_Photo	File stems of the two views	Photo numbers ≠ sherd numbers (see 5.2)
Site	Jinan_Daxinzhuang	Constant; one site
Context_Unit	Pit and layer, e.g. H690_Layer_8	43 units: stratigraphy, a potential prior (joining sherds usually lie in nearby layers)
Small_Find_No	Registered small-find number	Empty for the main sheet
Notes	Condition notes, e.g. Exterior_Spalled	Surface spalling changes appearance and explains some errors

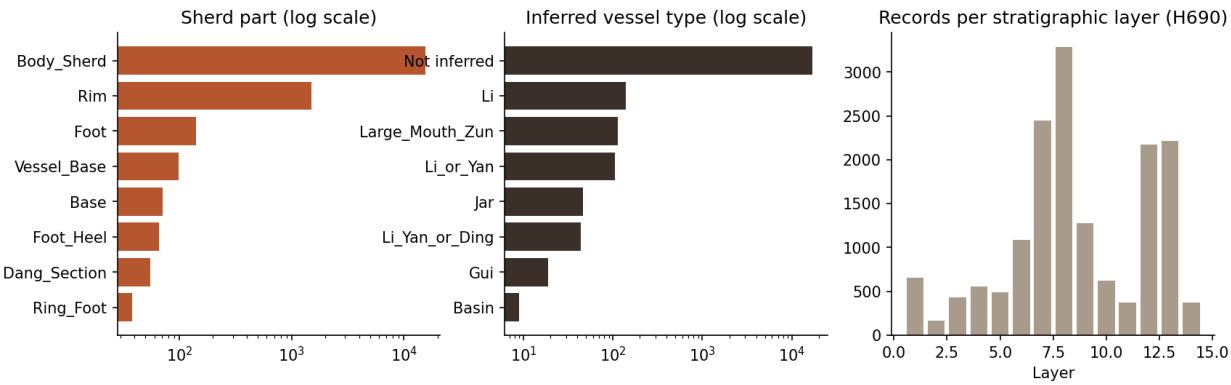


Figure 4. Metadata distributions (log scales): sherd parts, inferred vessel types (mostly not inferred), and records per stratigraphic layer of pit H690.

5.2 A metadata pitfall we found and fixed

A naive reading suggests that sherd JD00076 is photographed in file JD_00076.JPG. The metadata shows otherwise: each sherd has *two* photo numbers (JD00001 → JD_00001 exterior and JD_00002 interior), so photo numbers run about twice as fast as sherd numbers. Our first download script mapped IDs by number and would have silently paired the wrong photos. The bug was caught by reading the metadata, and the fixed script maps every refit sherd through the *Exterior_Photo* column. This is a concrete case of metadata preventing a label error that pixels alone could never reveal.

5.3 Annotation metadata (gt_all.json) and pose conventions

Each of the 111 gold records has *pair_id*, *side* (all exterior), *img_a*, *img_b* and two poses A_action, B_action = [x, y, cos θ, sin θ]. The README does not state the exact convention. From the dataset's reference script we established that each image is first recentred on its alpha-weighted centroid c, then rotated by θ and placed at (x, y), so world = R(θ)(p − c) + t. We confirmed this independently: under this convention the true partners touch along 75–170 px of edge with ≤7% overlap, whereas the alternative rotation sign gives large overlaps. The simulated data use a *different* convention: rotation about the image centre with the opposite sign of θ. We found it by testing all eight combinations of sign, centre and inversion and keeping the one that produced contact without overlap (211 px mean contact, 4% overlap). Figure 6 re-renders gold joins with our own implementation of the convention.

5.4 Why metadata matters for evaluation

- **Leakage control.** The vessel-group structure (derived from the join graph) defines the cross-validation folds, so no vessel appears in both training and test data.
- **Interpretation of errors.** Condition notes (spalling, adhered soil, residue) explain why some sherds' appearance does not match their partners.
- **Future priors.** Context unit and sherd part are not used by the model yet, but they are realistic priors (Section 14).

Table 8. First records of the Shandong sheet (Site = Jinan_Daxinzhuang for all).

Sherd_ID	Sherd_Part	Exterior_Photo	Interior_Photo	Context_Unit
JD00001	Rim	JD_00001	JD_00002	H690_Layer_1
JD00002	Rim	JD_00003	JD_00004	H690_Layer_1
JD00003	Body_Sherd	JD_00005	JD_00006	H690_Layer_1

6. Data Exploration and Preprocessing

Describe data quality issues and the preprocessing steps performed. Explain why each step is suitable for the data, such as handling missing values, duplicates, noise, outliers, scaling, encoding, segmentation, feature extraction or data augmentation. Include the sequence of steps.

Response:

6.1 Data quality issues found

Table 9. Data quality issues and how each is handled.

Issue	Where	Consequence if ignored	Handling
Background	Raw photos: grey/white backdrop + colour card + ink labels	Card or backdrop treated as sherd	Otsu on colour distance to the border colour, rectangularity filter for the card, GrabCut refinement
Illumination / colour cast	All images	Texture features follow lighting, not surface	CLAHE on the L channel (texture only); colour features use raw Lab
Arbitrary rotation	All images	Rotation-dependent features miss partners	Rotation-invariant LBP; 4-rotation averaged deep embeddings; geometry searches over all rotations
Scale must be preserved	Gold and sim	Per-image resizing would change edge lengths	One global working scale (0.5), never per-image resizing
Eroded / chipped edges	Real sherds	Edges do not fit exactly	Contour smoothing (σ = 1.5 samples), contact tolerance of 4 px, ICP refinement
Tiny specks / holes in the mask	Alpha matting edges	Spurious contours	Largest connected component, hole filling, morphological opening
Corrupt / partial files	Downloads	Crashes or silent bad data	CRC-32 check on extraction; image validation on load
Class imbalance (1.46%)	Pair labels	Classifier predicts 'no join' always	Class-balanced weights; PR-based metrics
Leakage via vessel	Pair labels	Inflated scores	Vessel-grouped folds

Missing values, duplicates and outliers in the usual tabular sense do not arise: every gold image has an annotation, no two images are identical (the largest DINOv2 cosine similarity between different sherds is well below 0.999), and every sherd area is plausible. Two of the 36,532 catalogue records lack an interior photo; this does not affect the exterior-only analysis.

6.2 Preprocessing sequence

Table 10. Preprocessing pipeline as implemented in src/pfm/preprocess.py.

Step	Operation	Parameters	Why it suits this data
01 Validate	Open and fully decode; reject empty masks	mask ≥ 200 px	Catches truncated downloads early
02 Standardise	Convert to RGBA; resize by a global factor	scale 0.5 (gold, sim); 1.0 (photos)	Keeps true relative size, which edge matching needs; halves compute
03 Isolate	Alpha > 127 (gold/sim) or Otsu + card filter + GrabCut (photos); largest component; fill holes	rectangularity < 0.88; GrabCut 4 iterations	Fragment shape must be exact for contour matching
04 Normalise	CLAHE on Lab L channel	clip 2.0, 8×8 tiles	Removes lighting gradients but keeps cord-mark relief
05 Edges	Outer contour, oriented consistently, resampled every 3 px, Gaussian-smoothed	step 3 px, σ = 1.5 samples	Equal spacing makes curvature windows comparable between sherds
06 Features	Colour/LBP histograms, deep embeddings, stripe orientation, curvature signature	see Table 6	Inputs for appearance and geometry cues

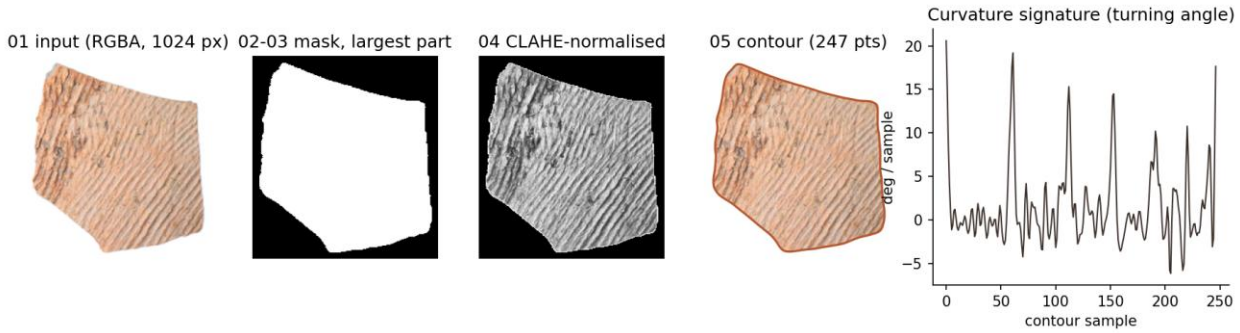


Figure 5. Preprocessing of sherd JDX-100: input, mask, CLAHE-normalised texture, resampled contour, and its curvature (turning-angle) signature. Peaks are corners of the outline.

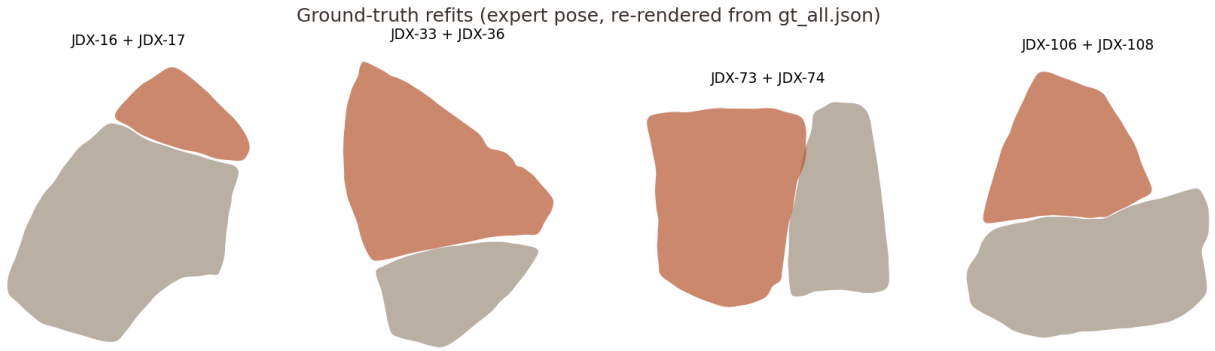


Figure 6. Gold-standard joins re-rendered from the expert poses with our own code. The pieces touch but leave gaps along eroded parts of the edge.

For raw excavation photos the isolation step is harder, because grey sherds sit on a grey backdrop next to a black-and-white scale card. Otsu thresholding alone picked the card in 2 of the first 6 test photos and under-segmented grey sherds. The rectangularity filter removes the card in every photo we inspected, and GrabCut recovers most of each sherd. Two errors remain (Figure 7): the light, low-contrast side of some sherds is cut off (JD_00028, JD_00076), and soft cast shadows are included on the other side. Both distort the outline, which matters for the planned raw-photo experiment (E6).

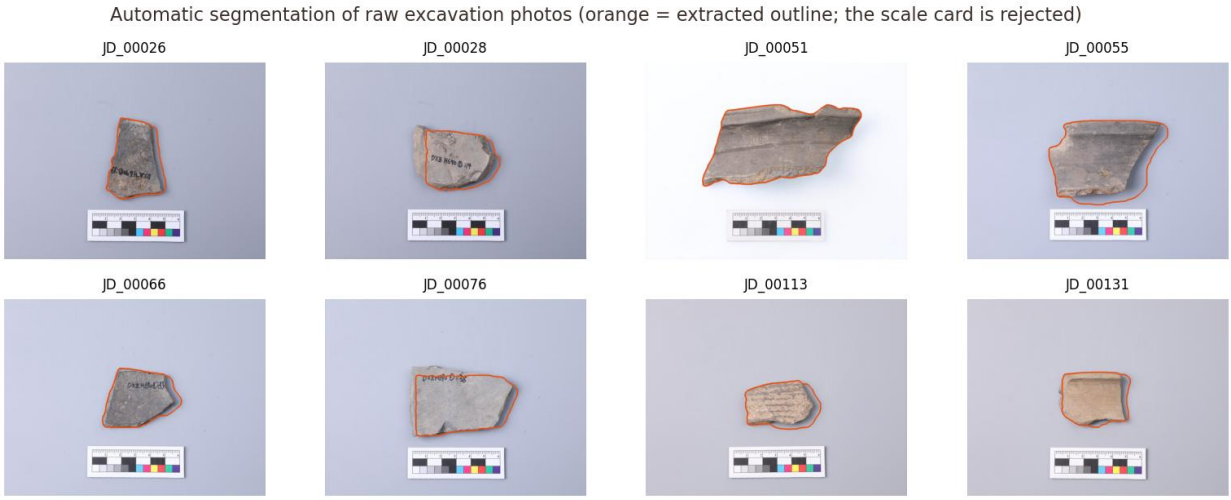


Figure 7. Automatic segmentation of raw excavation photos. The scale card is always rejected; light sherd edges are partly lost and shadows partly included.

7. Scope, Applications, Impact and Limitations

Explain the intended scope, practical applications, expected impact, assumptions and limitations of the proposed work.

Response:

Scope

The system ranks candidate joins for 2-D photographs of individual sherds and previews the best rigid alignment of a pair. It works on the exterior view, assumes sherds are roughly flat in the photo (2-D rigid motion), and is evaluated on one site's ware. It does **not** perform full multi-piece vessel reconstruction, and it does not decide joins. It produces a short list for an expert to check.

Applications

- Triage in finds processing: show the archaeologist the 5–10 most likely partners for each new sherd.
- Re-examining museum stores where sherds of one vessel ended up in different bags or layers.
- Digital archives: linking photographs of sherds across collections, without handling the objects.
- Teaching: the demo makes the difference between 'looks alike' and 'fits together' tangible.

Expected impact

With the proposed model, an expert who inspects the top 10 candidates finds a true partner for 71.8% of sherds, against 10.5% for random inspection of 10 sherds. In a collection of thousands of sherds, that means looking at ten candidates rather than at everything, for most sherds. The cost is a few seconds of computation per query.

Assumptions

- Photographs are taken at a consistent scale (true for the gold set; approximately true for the field photos, which include a scale card).
- Sherds are rigid and their outline is visible; joining edges are part of the outer contour.
- Expert joins are correct and complete for the pairs listed. Unlisted pairs are treated as non-joins, which can only make the measured precision pessimistic.

Limitations (summary; details in Section 14)

- Real fracture edges are eroded, which limits what 2-D geometry can recover.
- Only 111 positive pairs: enough to fit a linear model, not enough to train a deep network for this task.
- One site and one ware type; generalisation to other assemblages is untested.

8. Novelty / Innovation and Contribution

Identify what is original in the problem formulation, dataset preparation, feature design, model adaptation, evaluation or application. Distinguish the team's contribution from existing approaches.

Response:

The individual building blocks (LBP, CNN embeddings, Kabsch alignment, ICP, logistic regression) are standard and are cited as such. The contribution of this project lies in how they are combined and, above all, in how they are evaluated:

Table 11. Contribution of this case study versus existing approaches.

Aspect	What is new in this work	Existing practice
Problem formulation	Joining as pair classification *and* retrieval, scored with archaeologist-centred Hit@k on real expert joins	Sherd *classification* by type or decoration
Evaluation protocol	Vessel-grouped CV (no vessel in train and test), paired bootstrap CIs, synthetic-vs-real comparison of the same matcher	Random splits; synthetic breaks only
Feature design	Complementary curvature windows (A forward vs B reversed and negated), overlap-verified alignment, and a cord-mark orientation-consistency cue	Shape descriptors per sherd, no physical-fit check
Dataset preparation	Pose conventions of gold and simulated sets derived and verified; photo-ID mapping error found and fixed; range-request extraction of only the needed files	Not documented
Analysis	Ablation of every cue, error analysis (same-vessel confusions, sherd size), honest negative result for H1	Report best number only
Prototype	Interactive demo with ranked candidates, feature evidence and a fit preview	Offline scripts

Individual contribution. This is a one-person project. All code in the repository (preprocessing, features, geometric matcher, experiments, figures, demo, this report) was written, run and analysed by the student, using the public libraries and pre-trained weights listed in Section 16.

9. Pattern Recognition Model Selection and Working

Name the selected model/algorithm. Explain its core working principle, input/output, important stages and why it is appropriate for this problem.

Response:

Selected model: a two-cue pairwise classifier. Appearance and fracture-edge geometry features are fused by an L2-regularised, class-balanced logistic regression (LR), whose output probability ranks candidate partners.

9.1 Input, output and stages

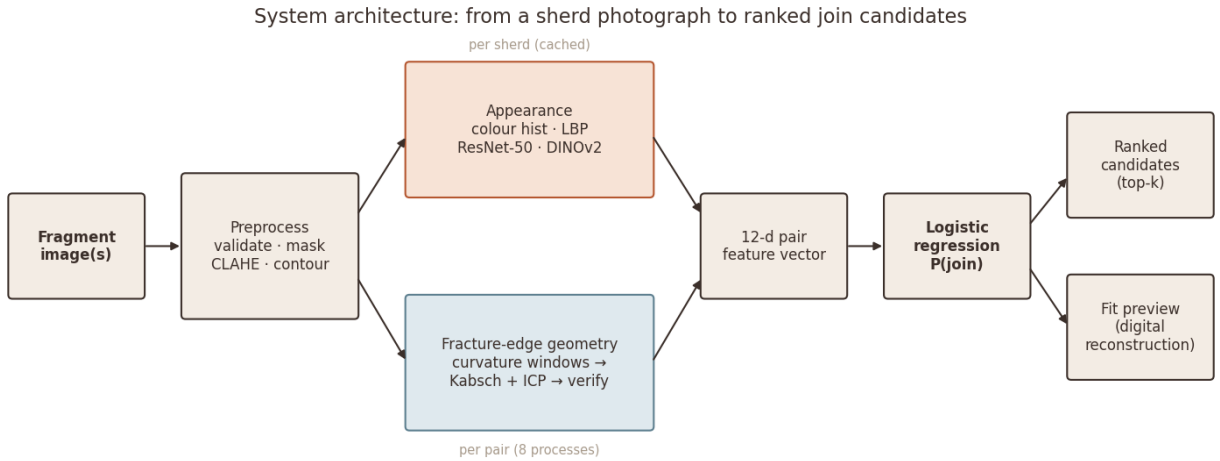


Figure 8. System architecture. Per-sherd descriptors are computed once and cached; the geometric matcher runs per pair.

Table 12. Processing stages.

Stage	Input	Output
1. Preprocess each sherd	Image file	Mask, normalised RGB, contour (N×2), centroid
2. Per-sherd descriptors	Preprocessed sherd	Colour hist (512), LBP (28), ResNet-50 (2048), DINOv2 (384), stripe orientation + coherence
3. Pair geometry	Two contours + orientations	Best rigid transform (R, t) and contact, overlap, rmse, orientation gap
4. Pair feature vector	Stages 2–3 for a pair	$x \in \mathbb{R}^{12}$ (Table 6)
5. Classifier	x	$P(\text{join} \mid x) = \sigma(w^T z + b)$, z = standardised x
6. Ranking	P(join) for query vs every gallery sherd	Ranked candidate list + fit preview of the top candidate

9.2 Fracture-edge geometric matcher (stage 3)

Where two sherds join, their outlines follow the same curve but are traversed in opposite directions, and a bump on one is a dent on the other. The matcher exploits this:

- **Curvature signature.** For each contour, the turning angle $\kappa_i = \text{wrap}(\angle(p_{i+1} - p_i) - \angle(p_i - p_{i-1}))$ is computed at every 3 px sample.
- **Complementary windows.** All windows of 32 samples (~96 px, stride 2) from A are compared with all windows from B *reversed and negated* ($-\kappa$_B read backwards), using squared Euclidean distance. Straight windows match everything, so the distance is divided by $(0.5 + \Sigma|\kappa|)$, favouring distinctive shapes.
- **Alignment.** For the 8 best seed pairs (with non-maximum suppression), the rigid transform mapping B's window onto A's is solved in closed form by the Kabsch/SVD method (Kabsch, 1976). It is then refined by two iterations of ICP (Besl & McKay, 1992) on all touching points.
- **Verification.** Each candidate pose is scored on the whole outline: contact = length of B's edge within 4 px of A; overlap = area of intersection / smaller area (solid sherds cannot overlap); score = contact × $\max(0, 1 - \text{overlap}/0.08)$ × $\exp(-(\Delta\varphi/20°)^2 \cdot c)$. Here $\Delta\varphi$ is the angle between the two sherds' cord-mark directions (structure-tensor estimate, Bigün & Granlund 1987) and c is its confidence. The best-scoring pose is kept.

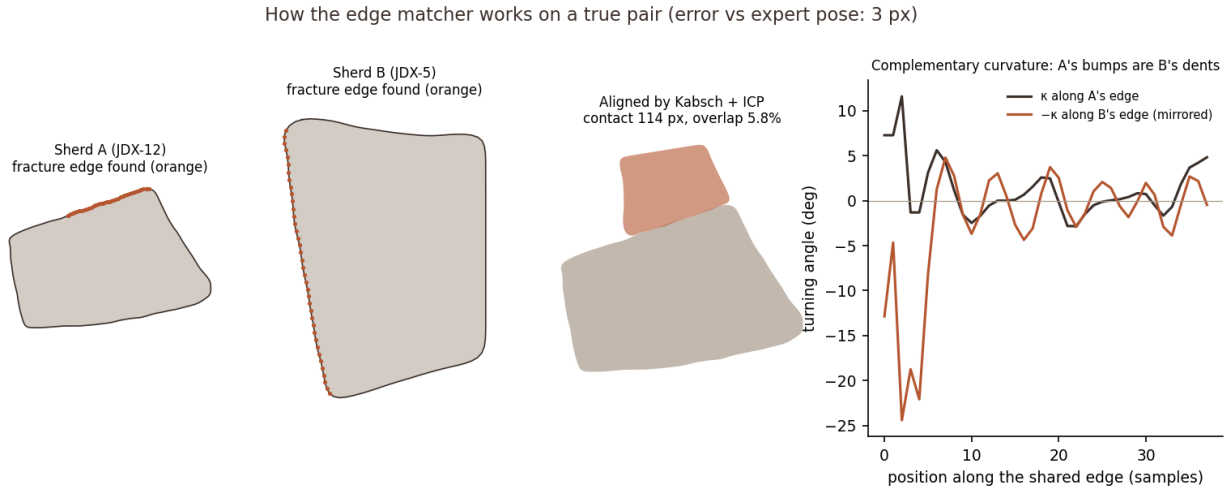


Figure 9. The matcher on a true pair (JDX-12 + JDX-5). It locates the shared fracture edge on both outlines (orange), aligns B to A, and the curvature of A along the joint mirrors that of B. The fit is checked against the whole outline, not only the seed window.

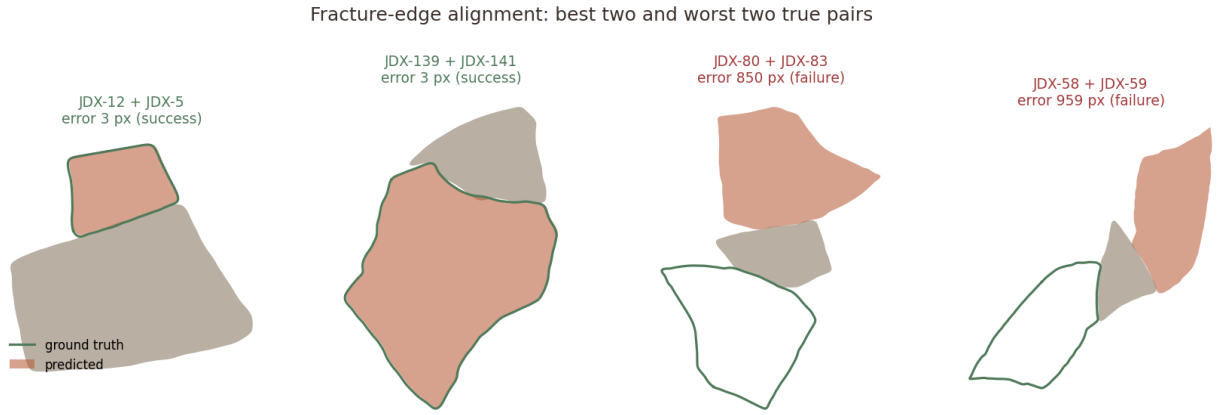


Figure 10. Output of the geometric matcher for true pairs: the two best (error 3 px, the predicted pose coincides with the expert pose) and the two worst (the matcher found a plausible-looking but wrong fit).

9.3 Logistic regression (stage 5)

LR models the log-odds of a join as a weighted sum of the standardised features: $\log[P/(1-P)] = b + \Sigma_j w_j z_j$. Weights are learned by minimising the class-weighted log-loss with an L2 penalty (C = 1). Class weights are inversely proportional to class frequency, so the 111 joins count as much as the 7,515 non-joins. At query time the predicted probability is computed for the query against every other sherd and sorted. Ranking only needs the order of the scores, so the decision threshold matters only for the pair-classification metrics, where it is chosen on training data (Section 11).

Algorithm 1: ranking the gallery for one query sherd q.

```
input : query image q, gallery G (cached descriptors), trained LR (w, b), scaler (μ, σ)
1  f_q ← preprocess(q)                      # mask, CLAHE, contour
2  d_q ← descriptors(f_q)                   # colour, LBP, ResNet-50, DINOv2, stripe angle
3  for each g in G:
4      a ← [−χ²(colour), −χ²(LBP), cos(ResNet), cos(DINOv2)]
5      (R, t) ← argmax over 8 curvature seeds of verify(Kabsch+ICP(f_q, f_g))
6      x ← [a, contact, contact_frac, overlap, rmse, score, Δϕ·c, c, |log area ratio|]
7      s_g ← σ( wᵀ (x − μ)/σ + b )               # P(join)
8  return G sorted by s_g (descending), with (R, t) of the top candidate for the fit preview
```

9.4 Why this model fits the problem

- The decision really is a weighing of evidence from different sources ("similar colour", "edges touch", "stripes line up"). LR expresses exactly that, and its weights can be read and checked for physical plausibility.
- With 111 positives, a model with 13 parameters is well determined. A deep network would overfit.
- It produces calibrated-ish probabilities, which are needed for a ranking across all candidates.
- Training takes milliseconds and prediction is a dot product, so the expensive part (geometry) is the only cost.

9.5 Working prototype

The model is packaged as an interactive Gradio application (app.py). The user either picks a sherd from the gold collection or uploads a new image: a PNG with a transparent background, or a photo on a plain backdrop, which is segmented automatically. They choose the proposed model or the image-only baseline. The app shows the top-5 candidates with their match scores, a table of the evidence behind each candidate (DINOv2 similarity, colour similarity, edge contact, overlap) and, for the best candidate, a preview of the physical fit, which is the first step toward digital reconstruction. For gallery sherds it also marks which candidates are expert-confirmed joins, so the model's behaviour can be checked live during the review. A gallery query answers instantly from cached features; an uploaded sherd takes about 2–6 s on a laptop CPU.

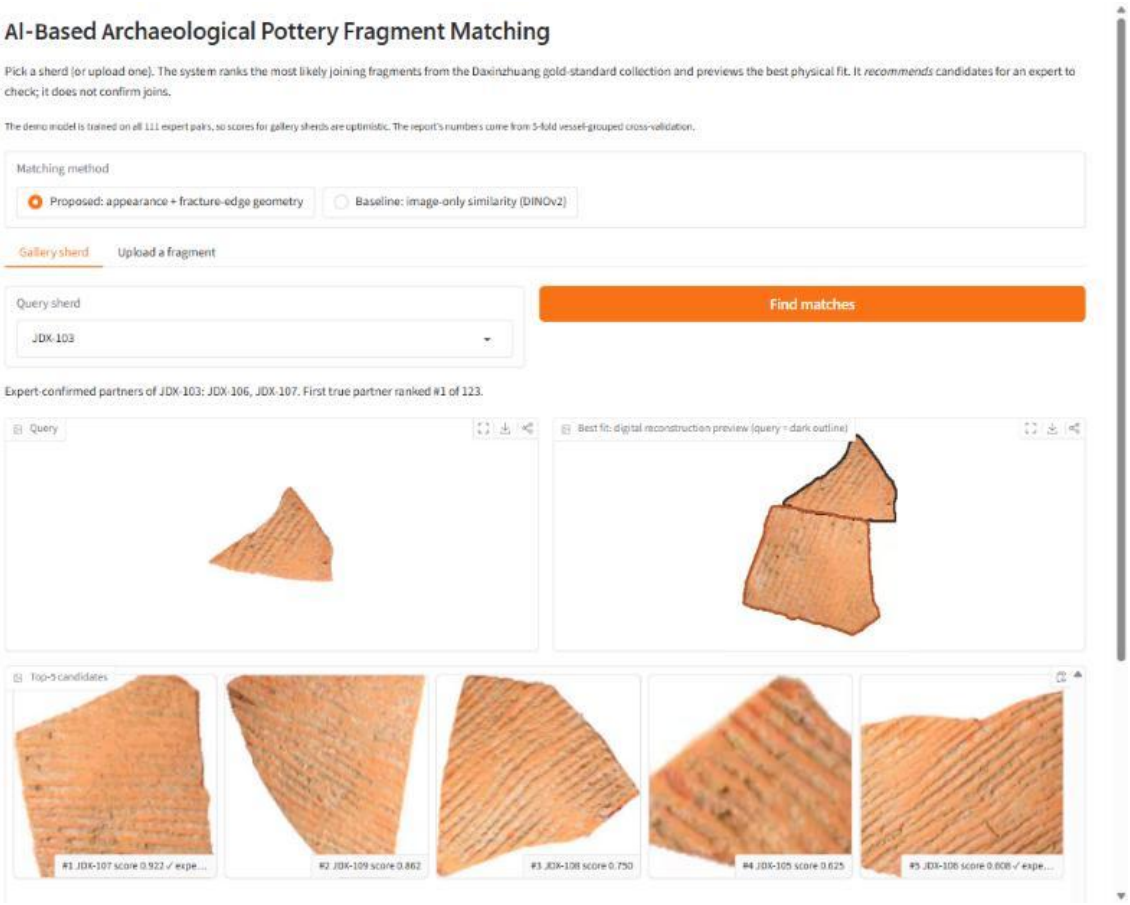


Figure 11. Demo: query JDX-103. Its expert-confirmed partner JDX-107 is ranked first, and the right panel previews the predicted fit.

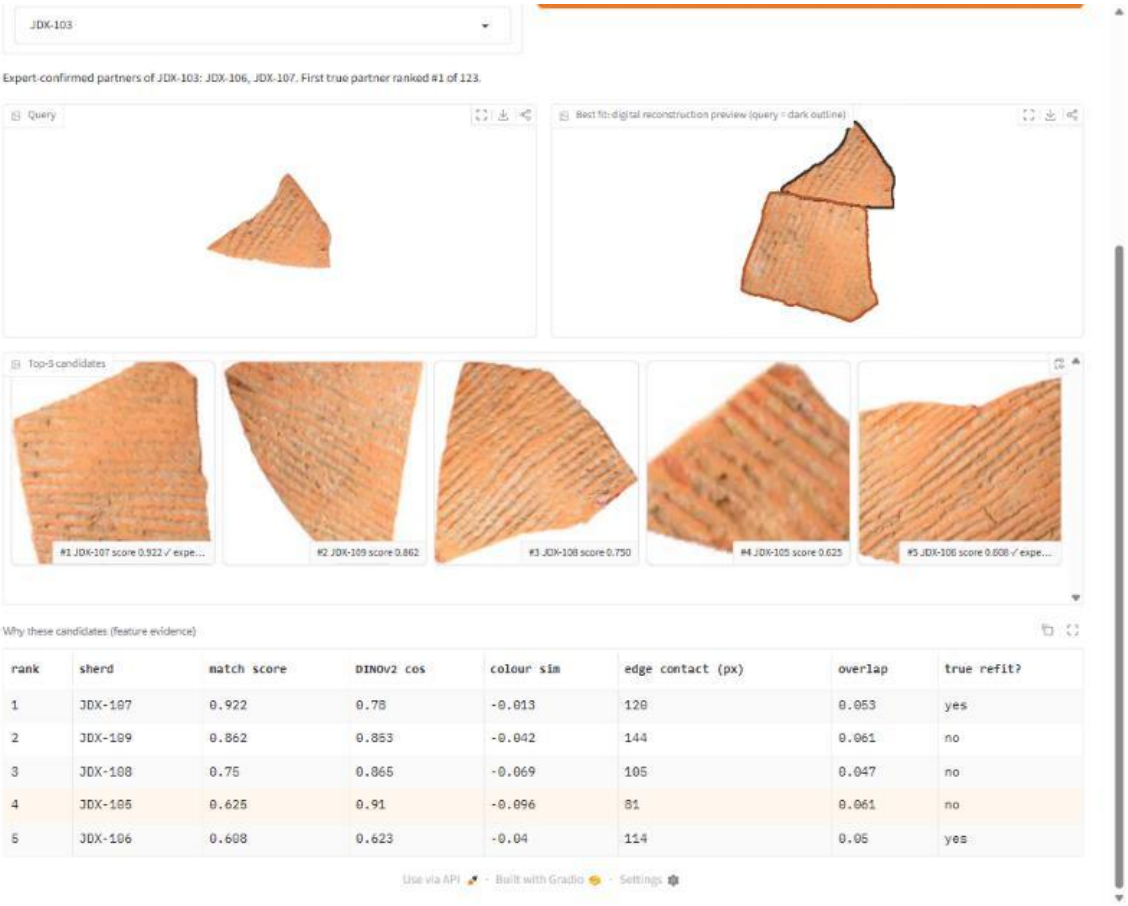


Figure 12. Demo: the evidence table explains each candidate's score, so the expert can see why a sherd was suggested.

Source code. The full project (pipeline, experiments, figures, demo and this report's generator) is public at https://github.com/vansh070605/pottery-fragment-matching, and every number in this report is regenerated by scripts/run_experiments.py and scripts/make_figures.py.

10. Justification for Model Selection

Give the reasons for selecting the model. Compare it with at least one suitable alternative using relevant criteria such as data characteristics, interpretability, complexity, computational cost, robustness and expected performance.

Response:

We compared the selected model with the image-only approach planned in Review 1 and with two non-linear classifiers trained on the same 12 features under the same folds:

Table 13. Model comparison on identical features and folds.

Model	Interpretability	Computational cost	Hit@5	Hit@10	MRR	Pair F1
DINOv2 embedding (cosine)	High (one similarity)	1 forward pass / sherd	0.403	0.540	0.314	–
Proposed: appearance + geometry (LR)	High (12 signed weights)	ms to train	0.556	0.718	0.364	0.287
Appearance + geometry (Random Forest)	Medium (feature importances)	seconds to train	0.484	0.645	0.340	0.291
Appearance + geometry (SVM-RBF)	Low (kernel)	seconds; scales $O(n^2)$	0.395	0.573	0.269	0.227

Table 14. Criteria-based comparison.

Criterion	Logistic regression (selected)	Random Forest	SVM (RBF)
Data characteristics: 12 continuous features, 111 positives	Few parameters, well-posed	Many trees on few positives, tends to memorise	Kernel width and C hard to tune with few positives

Criterion	Logistic regression (selected)	Random Forest	SVM (RBF)
Interpretability	Signed weights per feature (Figure 17)	Importances only, no direction	None
Complexity / cost	Linear; trivial	400 trees; moderate	Kernel matrix; grows quadratically
Robustness to imbalance	Class weights	Balanced subsampling	Class weights
Measured ranking quality	Best Hit@5/10 (0.556/0.718)	Lower ranking (0.484/0.645), similar F1	Lowest (0.395/0.573)

Against the **image-only baseline** (DINOv2 cosine, the Review 1 plan), the selected model improves Hit@5 by +15.3 pts (95% CI +5.6 to +25.0) and Hit@10 by +17.7 pts (95% CI +9.7 to +26.6). Against the **Random Forest**, it ranks better (Hit@5 0.556 vs 0.484) with a comparable pair F1 (0.287 vs 0.291). The forest trades precision for recall at its threshold but orders candidates less well. The **SVM** is worst on every metric. With so few positives the flexibility of non-linear models does not pay off, and the linear model's interpretability is a decisive advantage for an archaeological tool whose suggestions must be explained.

Why not an end-to-end deep (Siamese) network? It would need thousands of real positive pairs. The 111 real joins are too few, and the 50,736 simulated pairs come from clean synthetic breaks of a few vessels. Section 12 shows that the geometry of real breaks differs from synthetic ones, so a network trained on simulation would face a large domain gap. We therefore use pre-trained embeddings as fixed features and learn only a small fusion model, which is the appropriate choice for this data size.

11. Experimental Setup and Evaluation Method

Describe the train/validation/test split or validation strategy, implementation environment, parameters, baseline (if any) and evaluation metrics. Explain why the selected metrics are appropriate.

Response:

11.1 Validation strategy

5-fold vessel-grouped cross-validation. The 33 vessel groups (connected components of the join graph) are shuffled (seed 0) and assigned to 5 folds. For test fold k, the model is trained only on pairs whose two sherds both lie outside fold k, and it scores every pair that involves a fold-k sherd. Each query is ranked against the full gallery of 123 other sherds, as in real use. A pair whose two sherds fall in different test folds receives the average of its two scores. Because no vessel is ever in both train and test, the model cannot succeed by memorising a vessel's colour.

Decision threshold. For the pair-classification metrics, the threshold that maximises F1 is chosen by an inner 3-fold grouped CV on the training pairs only, then applied to the held-out pairs fully inside the test fold (1,542 pairs in total, 111 positive).

11.2 Experiments

Table 15. Experiments.

ID	Question	Data	Protocol
E1	Does fusion beat image-only retrieval?	Gold, 124 queries	Vessel-grouped 5-fold CV
E2	Which classifier?	Same	LR vs RF vs SVM-RBF on identical features and folds
E3	Which cues matter?	Same	Drop one cue group at a time from the proposed model
E4	Does the edge matcher work on clean breaks?	sim_data_8, 300 random pairs + 300 negatives	Pose error vs ground truth; AUC vs pairs from different fracture runs
E5	Does it recover real poses?	111 gold joins	Mean contour displacement vs expert pose
E6	Does it transfer to raw field photos?	Expert refit spreadsheet + distractor photos	Planned: gold-trained models, appearance top-30 then geometry re-rank (script ready, not run for this review)

11.3 Environment and parameters

Table 16. Implementation environment and parameters.

Item	Setting
Hardware / OS	Laptop, Windows 11; inference on CPU (PyTorch CPU build), 8 worker processes for geometry
Software	Python 3.12, PyTorch 2.9.1, timm 1.0.29, scikit-learn 1.4.1, OpenCV 4.11, scikit-image 0.26, Gradio 6

Item	Setting
Pre-trained models	resnet50.a1_in1k (ImageNet-1k); vit_small_patch14_dinov2.lvd142m at 224 px; 4 rotations averaged
Preprocessing	scale 0.5, contour step 3 px, smoothing σ 1.5, CLAHE clip 2.0
Geometry	window 32, stride 2, 8 seeds, contact tolerance 4 px, overlap penalty 0.08, orientation σ 20°
Classifiers	LR: C = 1, balanced; RF: 400 trees, min leaf 5, balanced subsample; SVM: RBF, C = 1, balanced
Randomness	All seeds = 0; results are deterministic
Runtime	Features for 124 sherds ≈ 40 s; geometry for all 7,626 pairs ≈ 1 min; full CV ≈ 1 min

11.4 Reproducibility

Table 17. Reproducing every result from the repository.

Step	Command	Output
Fetch photo subset	python scripts/fetch_real_subset.py	data/raw/Daxinzhuang-18K/… (refit + distractor photos)
Explore data	python scripts/explore_dataset.py	outputs/dataset_summary.json, dataset figures
E1–E5	python scripts/run_experiments.py	outputs/results.json, tables, trained model
E6 (planned)	python scripts/run_real_photos.py	outputs/results_real_photos.json
Figures + CIs	python scripts/make_figures.py; make_method_figures.py	outputs/figures, bootstrap_ci.csv
This report	python scripts/build_report.py	docs/Review2_Case_Study_Report.docx
Demo	python app.py	http://127.0.0.1:7860

11.5 Baselines and metrics

Baselines: random ranking; colour-histogram χ^2; LBP χ^2; ResNet-50 cosine; DINOv2 cosine (the Review 1 image-only plan); and the raw geometry score. Metrics and why they were chosen:

- **Hit@k** (k = 1, 5, 10): fraction of queries with at least one true partner in the top k. It is exactly what an archaeologist experiences when checking k candidates.
- **MRR**: mean of 1/rank of the first true partner. It rewards putting a partner at the very top.
- **mAP**: mean average precision over all partners of a query, for sherds with several partners.
- **CMC curve**: Hit@k for k = 1…30, showing the whole trade-off.
- **Precision, recall, F1, ROC-AUC, PR-AUC, confusion matrix** for the pair classifier. Accuracy is reported but is not informative: with 1.46% positives, predicting 'no join' for everything already scores 98.5%.
- **Paired bootstrap 95% CIs** (5,000 resamples of the 124 queries) for differences between methods, because 124 queries is a small test set.

12. Results and Analysis

Present results using suitable tables/graphs and, where applicable, a confusion matrix. Interpret precision, recall, F1-score, accuracy or other relevant metrics. Identify class-wise patterns, errors, misclassifications and possible reasons. Do not report accuracy alone.

Response:

12.1 E1: retrieval of join partners

Table 18. Retrieval results (124 queries, 123 candidates each, vessel-grouped CV). Higher is better except median rank.

Method	Hit@1	Hit@5	Hit@10	MRR	mAP	Med. rank
Random ranking	0.008	0.040	0.105	0.056	0.045	43.5
Colour histogram (chi2)	0.137	0.403	0.621	0.271	0.217	7.0
LBP texture (chi2)	0.113	0.274	0.419	0.210	0.147	13.5
ResNet-50 embedding (cosine)	0.153	0.419	0.565	0.276	0.206	8.5
DINOv2 embedding (cosine)	0.194	0.403	0.540	0.314	0.238	8.5
Geometry score only (no learning)	0.016	0.137	0.274	0.096	0.074	26.0
Geometry features (LR)	0.024	0.105	0.250	0.095	0.075	28.0
Appearance features (LR)	0.258	0.524	0.677	0.392	0.308	5.0
Proposed: appearance + geometry (LR)	0.194	0.556	0.718	0.364	0.288	4.0

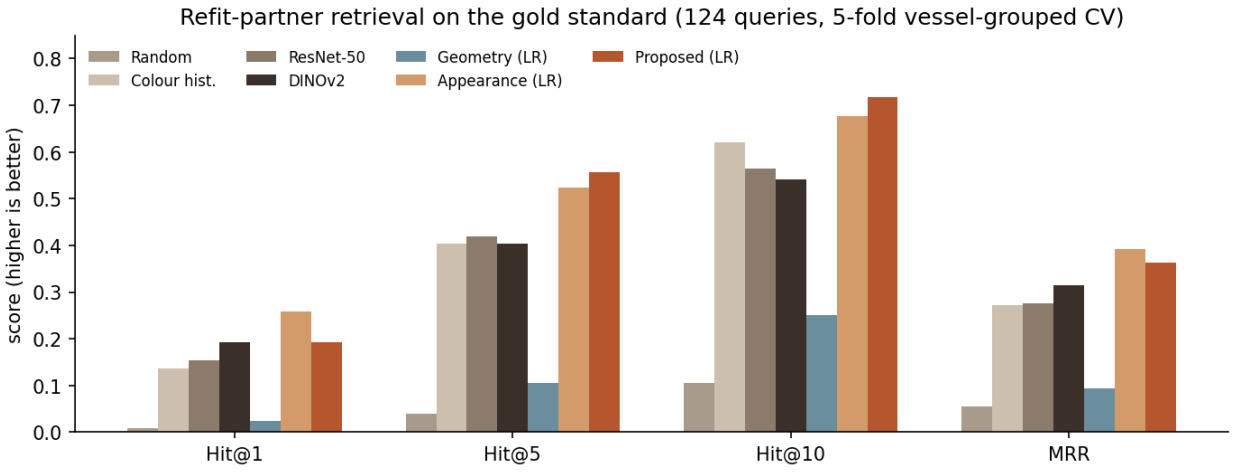


Figure 13. Retrieval metrics of the main methods.

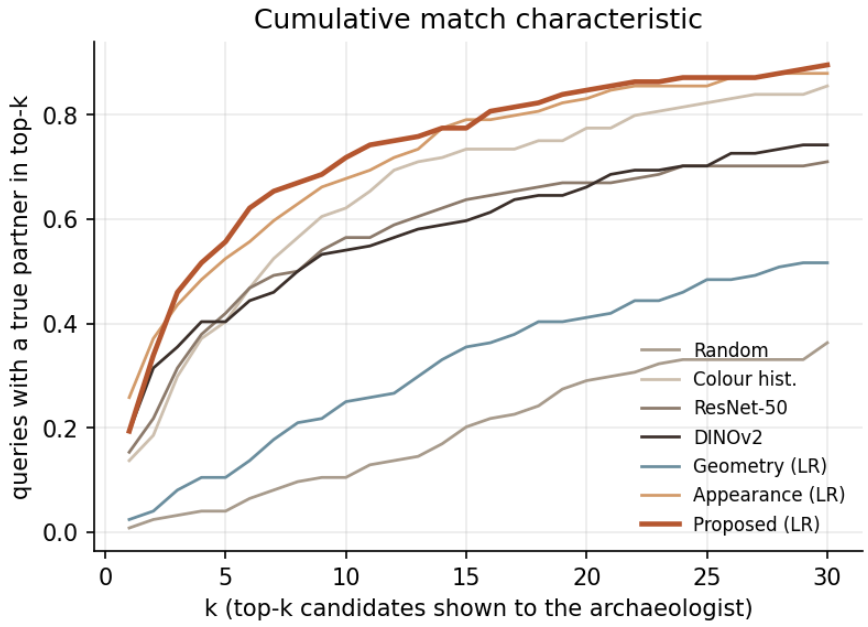


Figure 14. Cumulative match characteristic: share of queries with a true partner within the top k.

Reading the results. Every learned or pre-trained cue is far above random (Hit@10 0.105). Among single cues, the simple **colour histogram is as good as deep embeddings** (Hit@10 0.621 vs 0.540 for DINOv2). Sherds of one vessel share their firing colour, while the deep models also respond to shape and texture that differ between neighbouring sherds. DINOv2 is the best single cue at the very top (Hit@1 0.194). **Geometry alone is weak** (Hit@10 0.250): with eroded edges, many wrong pairs find an equally good-looking fit. **Learning to combine appearance cues** lifts Hit@5 from 0.403 to 0.524 and gives the best Hit@1 and MRR. **Adding geometry** gives the best Hit@5 (0.556), Hit@10 (0.718) and median rank (4), but lowers Hit@1 from 0.258 to 0.194.

12.2 Statistical significance

Table 19. Paired bootstrap differences in percentage points with 95% confidence intervals. An interval that excludes 0 is significant.

Difference	Hit@1	Hit@5	Hit@10	MRR
Proposed − DINOv2 (image-only)	+0.0 pts [-8.1 to +7.3]	+15.3 pts [+5.6 to +25.0]	+17.7 pts [+9.7 to +26.6]	+5.0 pts [-0.9 to +11.3]
Proposed − Appearance LR	-6.5 pts [-12.1 to -1.6]	+3.2 pts [-2.4 to +9.7]	+4.0 pts [-2.4 to +10.5]	-2.8 pts [-6.7 to +0.9]
Appearance LR − DINOv2	+6.5 pts [-2.4 to +14.5]	+12.1 pts [+2.4 to +21.8]	+13.7 pts [+4.8 to +22.6]	+7.8 pts [+1.3 to +14.8]

The improvement over the image-only baseline at Hit@5 and Hit@10 is significant. The improvement from adding geometry on top of learned appearance is **not** significant at Hit@5/10, and the drop at Hit@1 is.

Table 20. Results per cross-validation fold.

Fold	Queries	Vessels	DINOv2 Hit@5	DINOv2 Hit@10	Appearance (LR) Hit@5	Appearance (LR) Hit@10	Proposed (LR) Hit@5	Proposed (LR) Hit@10
1	29	7	0.517	0.655	0.586	0.793	0.621	0.724
2	25	7	0.240	0.440	0.480	0.600	0.600	0.720
3	29	7	0.448	0.483	0.448	0.586	0.379	0.724
4	15	6	0.333	0.533	0.267	0.533	0.267	0.467
5	26	6	0.423	0.577	0.731	0.808	0.808	0.846
Std. dev.			0.108	0.083	0.172	0.127	0.213	0.139

The per-fold view shows how much results depend on which vessels are held out. The proposed model beats the image-only baseline at Hit@10 in 4 of 5 folds and at Hit@5 in 3 of 5. Fold-to-fold spread of about ±14 points is comparable to the differences between methods. This is one more reason to rely on the pooled bootstrap intervals rather than on any single split, and it shows that a larger test set is needed before smaller effects can be resolved.

12.3 Pair classification and confusion matrix

Table 21. Pair classification on held-out pairs (1,542 pairs, 111 joins; chance PR-AUC = 0.072).

Model	Precision	Recall	F1	ROC-AUC	PR-AUC	Accuracy	TP/FP/FN/TN
Geometry features (LR)	0.139	0.081	0.102	0.592	0.108	0.897	9/56/102/1375
Appearance features (LR)	0.411	0.207	0.275	0.827	0.252	0.921	23/33/88/1398
Proposed: appearance + geometry (LR)	0.371	0.234	0.287	0.825	0.247	0.916	26/44/85/1387
Appearance + geometry (Random Forest)	0.294	0.288	0.291	0.794	0.237	0.899	32/77/79/1354
Appearance + geometry (SVM-RBF)	0.250	0.207	0.227	0.786	0.228	0.898	23/69/88/1362

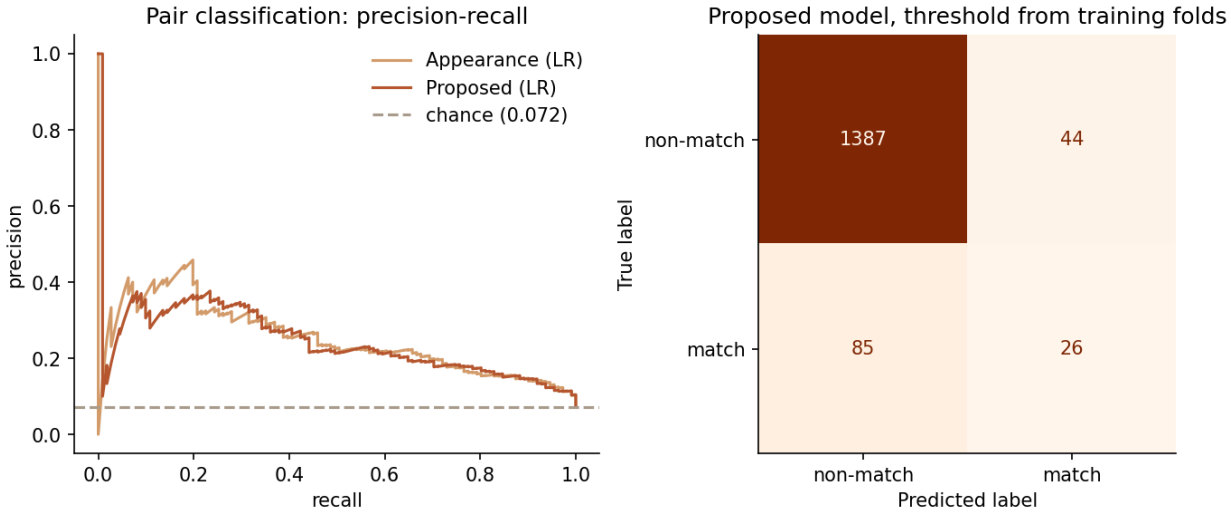


Figure 15. Precision-recall curves and the confusion matrix of the proposed model at the threshold chosen on training folds.

At its operating point the proposed model flags 70 pairs as joins, of which 26 are correct (precision 0.371), and finds 26 of the 111 joins (recall 0.234). Accuracy is 91.6%, but it is almost meaningless here: the trivial all-'no join' classifier reaches 92.8% on these test pairs. PR-AUC is 0.247, 3.4× the chance level. ROC-AUC is high (0.825) because the model easily rejects obviously different sherds. The hard part is the small set of look-alike non-joins, which the PR curve exposes and the ROC curve hides. As a stand-alone verdict ('these two join') the model is not reliable. As a ranker that shortens the list it is useful, which matches its intended use.

12.4 E3: ablation and model weights

Table 22. Ablation: each row removes one group of features from the proposed model.

Variant	Hit@1	Hit@5	Hit@10	MRR	Pair F1
Full proposed model	0.194	0.556	0.718	0.364	0.287
without orientation cue	0.218	0.589	0.718	0.379	0.324
without overlap check	0.185	0.556	0.702	0.357	0.293
without edge contact	0.202	0.589	0.742	0.365	0.303
without deep embeddings	0.121	0.427	0.645	0.271	0.246
without colour	0.161	0.435	0.556	0.293	0.273

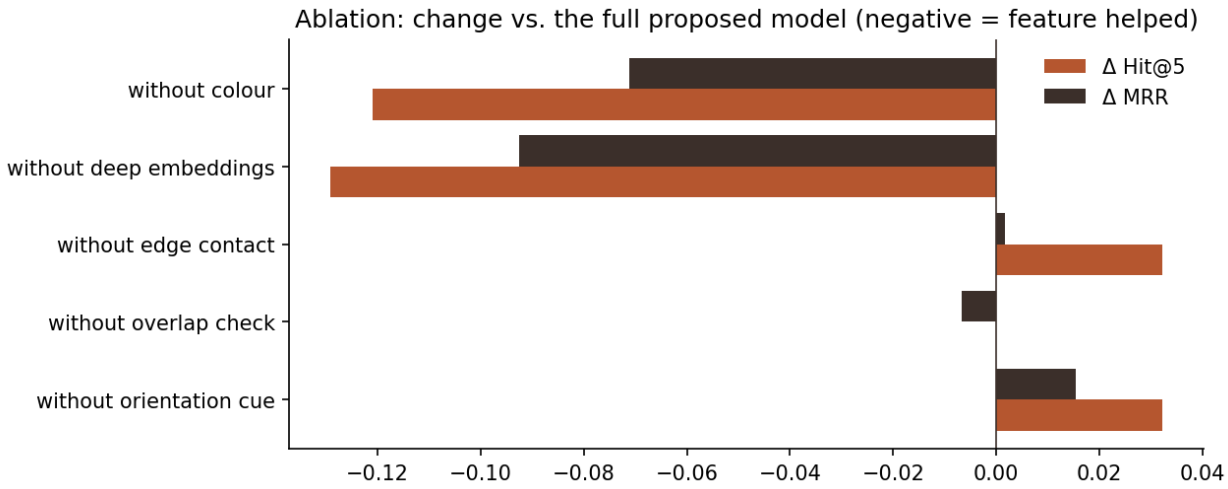


Figure 16. Change in Hit@5 and MRR when a cue is removed. Negative bars mean the cue helped.

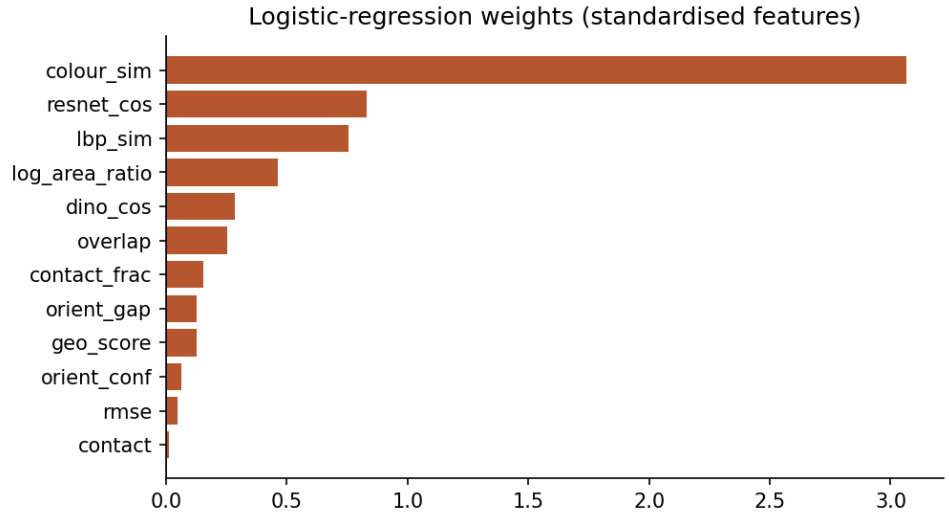


Figure 17. Standardised logistic-regression weights of the proposed model (trained on all gold pairs).

Removing **colour** or the **deep embeddings** causes the largest drops (Hit@5 to 0.435 and 0.427), confirming that appearance carries the signal. Removing the **orientation cue** *improves* every metric (Hit@5 0.589, F1 0.324), and removing the edge-contact features improves Hit@5/10. The geometric features therefore add noise more than information on these sherds. We did **not** switch the headline model to the best ablation: choosing a model by its test-fold performance would bias the result upward. Instead we report the pre-specified model and treat the ablation as a finding. The LR weights tell the same story: colour (+3.07) and ResNet (+0.83) dominate. Some geometric weights even have physically implausible signs: overlap is *positively* weighted (+0.25) and so is the orientation gap (+0.13). This is evidence that, at mostly wrong poses, these features capture incidental shape correlations rather than real fit.

12.5 E4 and E5: geometry on synthetic versus real breaks

Table 23. Pose recovery of the same geometric matcher.

Data	Pairs	Pose within 50 px	Pose within 100 px	Median rotation error
Synthetic clean breaks (sim_data_8)	300	47.0%	50.0%	3.3°
Real eroded sherds (gold)	111	17.1%	20.7%	69.7°

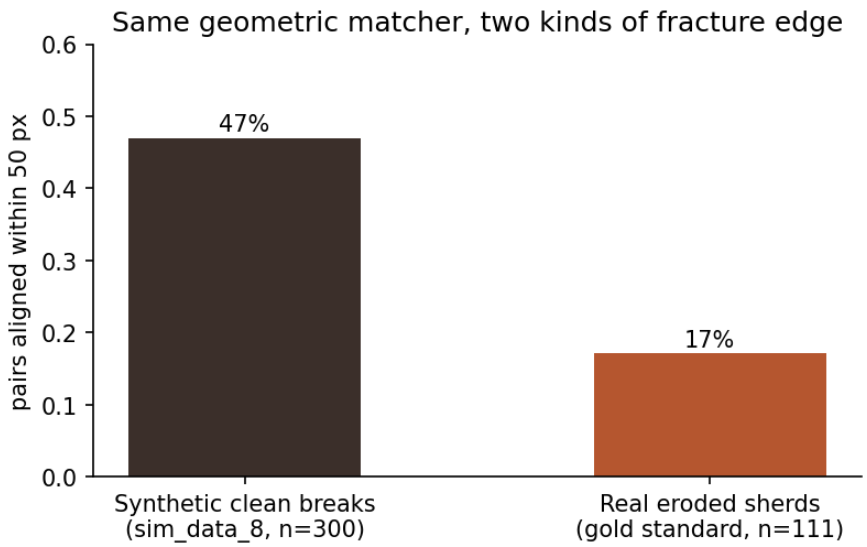


Figure 18. The geometric matcher recovers the correct joint for about half of the clean synthetic breaks but under a fifth of the real ones.

On synthetic breaks the matcher finds the exact joint for 47.0% of pairs (median rotation error 3.3°). On real sherds this falls to 17.1% (median rotation error 70°). A diagnostic on 40 real pairs showed why. At the *expert* pose, true partners touch along a median of 87 px of edge, whereas the best *wrong* pose that ICP finds touches along 129 px. On worn edges, maximising contact actively prefers wrong fits. Even on synthetic data, the geometry score separates joins from non-joins only modestly (AUC 0.590), because the negatives come from the same vessel and straight break segments fit many places.

12.6 Error analysis

- **Wrong top-1 candidates are mostly other vessels.** For the proposed model, 100 queries have a wrong top-1, and only 11 of those (11.0%) are sherds of the same vessel that simply do not touch the query. The main confusion is therefore between *different vessels of the same ware*, not between neighbours within one vessel.
- **The vessel is still found.** 21.9% of the top-5 candidates belong to the query's vessel, against 3.2% expected by chance, a 7× enrichment. So the model groups sherds by vessel even when it misses the exact neighbour.
- **Small sherds are hardest.** Hit@5 rises from 0.439 (smallest third of sherds) to 0.714 (largest third). Small sherds have less surface and shorter edges. The proposed model helps them most: DINOv2 alone reaches only 0.244 on the smallest third.
- **Larger vessels are easier** (Hit@5 0.625 for vessels of ≥ 5 sherds vs 0.523 for ≤ 3), because their queries have more partners to find.
- **Spalled surfaces and colour variation within a vessel** (one side of a pot fired darker) explain failures such as JDX-118 in Figure 19. Its partners differ in tone, and the retrieved candidates are plausible look-alikes from other vessels.

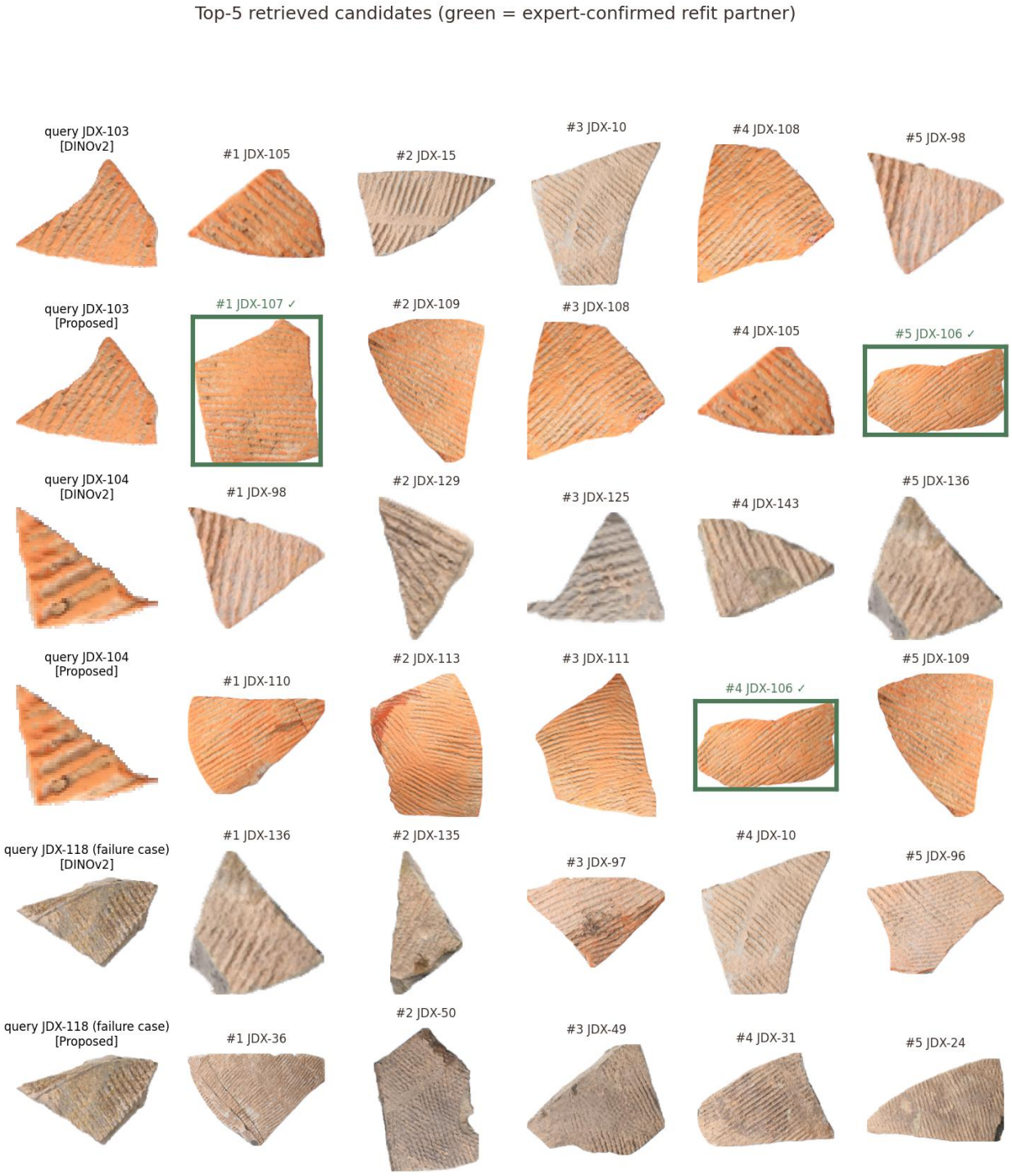


Figure 19. Top-5 candidates from the image-only baseline and the proposed model for three queries (green = expert-confirmed partner). In the first two the proposed model finds partners the baseline misses; the third is a failure case for both.

12.7 Complete results of all configurations

Table 24. All 16 configurations. 'Vessel mAP' scores retrieval of any sherd of the same vessel instead of direct neighbours only.

Configuration	Family	Hit@1	Hit@5	Hit@10	MRR	mAP	Vessel mAP	Pair F1
Random ranking	baseline	0.008	0.040	0.105	0.056	0.045	0.059	–
Colour histogram (chi2)	baseline	0.137	0.403	0.621	0.271	0.217	0.223	–
LBP texture (chi2)	baseline	0.113	0.274	0.419	0.210	0.147	0.174	–
ResNet-50 embedding (cosine)	baseline	0.153	0.419	0.565	0.276	0.206	0.230	–
DINOv2 embedding (cosine)	baseline	0.194	0.403	0.540	0.314	0.238	0.266	–

Configuration	Family	Hit@1	Hit@5	Hit@10	MRR	mAP	Vessel mAP	Pair F1
Geometry score only (no learning)	baseline	0.016	0.137	0.274	0.096	0.074	0.085	–
Geometry features (LR)	learned	0.024	0.105	0.250	0.095	0.075	0.089	0.102
Appearance features (LR)	learned	0.258	0.524	0.677	0.392	0.308	0.308	0.275
Proposed: appearance + geometry (LR)	proposed	0.194	0.556	0.718	0.364	0.288	0.303	0.287
Appearance + geometry (Random Forest)	alternative	0.202	0.484	0.645	0.340	0.271	0.288	0.291
Appearance + geometry (SVM-RBF)	alternative	0.137	0.395	0.573	0.269	0.204	0.241	0.227
Ablation: proposed - orientation cue	ablation	0.218	0.589	0.718	0.379	0.295	0.305	0.324
Ablation: proposed - overlap check	ablation	0.185	0.556	0.702	0.357	0.289	0.303	0.293
Ablation: proposed - edge contact	ablation	0.202	0.589	0.742	0.365	0.290	0.302	0.303
Ablation: proposed - deep embeddings	ablation	0.121	0.427	0.645	0.271	0.216	0.227	0.246
Ablation: proposed - colour	ablation	0.161	0.435	0.556	0.293	0.223	0.251	0.273

13. Inference and Interpretation

State what can be concluded from the evidence. Explain what the model learned or failed to learn, what the errors indicate and whether the objectives were achieved. Avoid unsupported claims.

Response:

What the evidence supports

- **A learned multi-cue ranker is a practical triage tool.** Under leak-free evaluation, an expert who checks 10 suggestions finds a true partner for 71.8% of sherds, a significant improvement over the image-only plan of Review 1 (54.0%).
- **Hypothesis H1 is only partly supported.** The full model is significantly better than the image-only *baseline*, but the improvement is explained by fusing appearance cues. Adding fracture-edge geometry to learned appearance gives no significant gain at top-5/10 and a significant loss at top-1.
- **The bottleneck is the edge data, not the matching algorithm.** The same matcher works on half of the clean breaks and under a fifth of the real ones. On eroded edges, a maximum-contact criterion prefers wrong fits.
- **Colour is the most informative cue for this ware.** It matches deep features as a single cue and has the largest model weight. Vessels were fired unevenly, so colour identifies the vessel better than shape does.

What the model learned and failed to learn

The model learned to *group sherds by vessel*: its top 5 is 7× enriched in same-vessel sherds. It did not learn to identify the *adjacent* sherd within a vessel, which is what geometry was meant to add. The physically implausible signs of some geometric weights (positive weight on overlap) show that the geometric features, measured at mostly wrong poses, do not carry the meaning they were designed for.

Were the objectives achieved?

Objectives O1–O7 were all met: the data are understood and documented, the pipeline runs end to end, the baseline and the proposed model were evaluated rigorously, alternatives were compared, and a working prototype exists. The research hypothesis received a clear, evidence-based answer, which is partly negative. We regard that as a valid outcome of the case study, not a failure of it.

Claims we do not make

- We do not claim the system reconstructs vessels. It ranks candidates and previews one rigid fit per pair.
- We do not claim that geometry is useless for sherd joining. Our 2-D outline-based geometry is not robust to erosion; 3-D or thickness-profile geometry was not tested.
- Results are for one site and ware. Other assemblages may favour different cues.

14. Limitations and Future Scope

Identify technical and data-related limitations. Suggest realistic improvements, additional data, alternative methods or future extensions.

Response:

Technical limitations

- 2-D outlines lose the third dimension: sherd thickness and wall curvature, which experts use first, are invisible.
- Maximum-contact pose selection is fragile on worn edges (Section 12.5). Only the outer contour is used, so a joining edge hidden by perspective is missed.
- The cord-mark orientation estimate is a single global direction. Sherds whose stripes curve or cross are poorly described.
- Pre-trained embeddings were not fine-tuned on pottery.
- Raw-photo segmentation sometimes includes cast shadows.

Data-related limitations

- 111 positive pairs is small: confidence intervals are wide (±5–10 points), and ablation differences of a few points are not significant.
- Only exterior views were used; the gold set also contains interior views.
- Unlisted pairs are assumed not to join; a few may be undiscovered joins.
- One site, one period, one dominant ware.

Future scope

Table 25. Realistic next steps.

Improvement	Expected benefit	Effort
Run E6: zero-shot transfer to 799 raw field photos (script and photos ready)	Tests the conclusions outside the curated benchmark	Low
Use the interior view as a second appearance cue	Independent evidence of the same vessel	Low
Edge-band appearance: compare colour/texture in a thin strip along the aligned joint	Checks continuity across the break, robust to erosion	Medium
Erosion-robust pose scoring learned from the gold poses (positive = expert pose, negative = ICP false fits)	Fixes the contact-maximisation failure	Medium
Fine-tune DINOv2 with a contrastive loss on same-vessel pairs (Yu et al. 2026 style)	Better vessel grouping	Medium
Context priors from metadata (layer distance, sherd part)	Down-weights implausible candidates	Low
Pre-train geometry on all 50,736 simulated pairs, adapt with augmentation that simulates erosion	Bridges the synthetic-real gap	High
Multi-piece assembly: chain pairwise fits into a vessel graph with global consistency	Toward full digital reconstruction	High
3-D scans or thickness profiles	Restores the strongest physical cue	High (needs new data)

15. Conclusion

Summarise the problem, approach, major findings, practical significance and key learning from the case study.

Response:

This case study turned the archaeological task of finding joining pottery sherds into a pattern-recognition problem, a pairwise join/no-join decision used to rank candidates, and evaluated it on real expert-confirmed joins from the Daxinzhuang-18K benchmark. We built a complete pipeline: segmentation and illumination normalisation; colour, texture and deep appearance features; a fracture-edge matcher based on complementary curvature, Kabsch/ICP alignment and overlap checks; a logistic-regression fusion model; and an interactive demo.

Under leak-free vessel-grouped cross-validation, the system puts a true partner in the top 10 for 71.8% of sherds, significantly above the image-only baseline planned in Review 1 (54.0%). The most important finding is less flattering and more useful: the gain comes from combining appearance cues, while 2-D fracture-edge geometry adds no significant benefit on real eroded sherds, even though the same matcher works on clean synthetic breaks. In practice, an appearance ranker can already save experts time today, whereas geometry needs erosion-robust descriptors before it can help.

Key learning: in pattern recognition, the evaluation protocol decides what a number means. Splitting by vessel, using ranking metrics instead of accuracy on a 1.5%-positive problem, bootstrapping the differences, and checking the method on synthetic data turned an apparently positive result into a precise statement of what works and why.

16. References

List datasets, research papers, books, software libraries and other sources in a consistent citation style.

Response:

[1] Ji, D., Xue, T., Wang, Q., Ma, W., Lang, J., Fang, H., Tu, C., & Lu, L. (2026). Daxinzhuang-18K: A large-scale 2D benchmark for pottery sherd retrieval and reassembly (Version 1.0.0) [Data set]. Zenodo. https://doi.org/10.5281/zenodo.21553799

[2] Freeman, H., & Garder, L. (1964). Apictorial jigsaw puzzles: The computer solution of a problem in pattern recognition. IEEE Transactions on Electronic Computers, EC-13(2), 118–127. https://doi.org/10.1109/PGEC.1964.263781

[3] Wolfson, H. J. (1990). On curve matching. IEEE Transactions on Pattern Analysis and Machine Intelligence, 12(5), 483–489. https://doi.org/10.1109/34.55108

[4] Papaioannou, G., Karabassi, E.-A., & Theoharis, T. (2002). Reconstruction of three-dimensional objects through matching of their parts. IEEE TPAMI, 24(1), 114–124. https://doi.org/10.1109/34.982888

[5] Huang, Q.-X., Flöry, S., Gelfand, N., Hofer, M., & Pottmann, H. (2006). Reassembling fractured objects by geometric matching. ACM Transactions on Graphics (SIGGRAPH), 25(3), 569–578. https://doi.org/10.1145/1179352.1141925

[6] Kampel, M., & Sablatnig, R. (2003). Profile-based pottery reconstruction. In Proc. CVPR Workshop on Applications of Computer Vision in Archaeology. https://doi.org/10.1109/CVPRW.2003.10007

[7] Chetouani, A., Treuillet, S., Exbrayat, M., & Jesset, S. (2020). Classification of engraved pottery sherds mixing deep-learning features by compact bilinear pooling. Pattern Recognition Letters, 131, 1–7. https://doi.org/10.1016/j.patrec.2019.12.009

[8] Pinho, M., Mota, G., & da Costa, G. (2025). A deep learning approach to assist in pottery reconstruction from its sherds. Heritage, 8(5), 167. https://doi.org/10.3390/heritage8050167

[9] Yu, X., Li, T., Song, Z., Lu, H., Du, Y., Wu, F., Xiao, H., & Xu, L. (2026). Mitigating spurious features by contrastive learning in pottery sherd recognition. npj Heritage Science, 14, 135. https://doi.org/10.1038/s40494-025-02170-3

[10] He, K., Zhang, X., Ren, S., & Sun, J. (2016). Deep residual learning for image recognition. In Proc. IEEE CVPR (pp. 770–778). https://doi.org/10.1109/CVPR.2016.90

[11] Oquab, M., Darcet, T., Moutakanni, T., et al. (2024). DINOv2: Learning robust visual features without supervision. Transactions on Machine Learning Research. arXiv:2304.07193

[12] Ojala, T., Pietikäinen, M., & Mäenpää, T. (2002). Multiresolution gray-scale and rotation invariant texture classification with local binary patterns. IEEE TPAMI, 24(7), 971–987. https://doi.org/10.1109/TPAMI.2002.1017623

[13] Kabsch, W. (1976). A solution for the best rotation to relate two sets of vectors. Acta Crystallographica Section A, 32(5), 922–923. https://doi.org/10.1107/S0567739476001873

[14] Besl, P. J., & McKay, N. D. (1992). A method for registration of 3-D shapes. IEEE TPAMI, 14(2), 239–256. https://doi.org/10.1109/34.121791

[15] Rother, C., Kolmogorov, V., & Blake, A. (2004). "GrabCut": Interactive foreground extraction using iterated graph cuts. ACM Transactions on Graphics, 23(3), 309–314. https://doi.org/10.1145/1015706.1015720

[16] Zuiderveld, K. (1994). Contrast limited adaptive histogram equalization. In P. Heckbert (Ed.), Graphics Gems IV (pp. 474–485). Academic Press.

[17] Bigün, J., & Granlund, G. H. (1987). Optimal orientation detection of linear symmetry. In Proc. 1st IEEE ICCV (pp. 433–438).

[18] Efron, B., & Tibshirani, R. J. (1993). An Introduction to the Bootstrap. Chapman & Hall.

[19] Pedregosa, F., et al. (2011). Scikit-learn: Machine learning in Python. Journal of Machine Learning Research, 12, 2825–2830.

[20] Paszke, A., et al. (2019). PyTorch: An imperative style, high-performance deep learning library. In Advances in Neural Information Processing Systems 32.

[21] Wightman, R. (2019). PyTorch Image Models (timm). GitHub repository. https://github.com/huggingface/pytorch-image-models

[22] Bradski, G. (2000). The OpenCV library. Dr. Dobb's Journal of Software Tools.

[23] Project code and results: Agrawal, V. (2026). pottery-fragment-matching. GitHub repository. https://github.com/vansh070605/pottery-fragment-matching

Review 1 – Problem, Data, Metadata & Preprocessing

Total: 7.5 Marks. This review assesses conceptual understanding of the problem, data and proposed approach—not merely project execution.

Rubric	Maximum Marks	Marks Awarded
Problem / Case Study Understanding	1.5	
Data Understanding	1.5	
Metadata Understanding	1.0	
Preprocessing	1.5	
Scope & Impact	1.0	
Novelty / Innovation	1.0	

TOTAL	**7.5**	

Faculty comments / probing questions: No faculty comments were recorded at Review 1.

Student's response / action items: Proceeded with the plan presented at Review 1 (dataset → preprocessing → image-only baseline → geometry-based matching). Review 2 implements and evaluates it; see Sections 9–14.

Review 2 – Model Selection, Results & Inference

Total: 7.5 Marks. This review assesses the student's ability to select, justify and critically evaluate the pattern recognition model and derive meaningful conclusions. High accuracy alone is not sufficient for high marks.

Rubric	Maximum Marks	Marks Awarded
Model Selection & Understanding	1.5	
Justification for Model Selection	2.0	
Results Analysis	1.5	
Inference / Interpretation	1.5	
Limitations & Future Scope	1.0	
TOTAL	**7.5**	

Faculty comments / probing questions:

_____

_____

Student's response / action items:

_____

_____

Performance-Level Reference

Level	Marks Range (out of 7.5)	Faculty Interpretation
Excellent	>7.0–7.5	Strong conceptual understanding; convincing justification; critical analysis; meaningful inferences; confident answers.
Good	5.5–7.0	Good understanding with minor gaps; generally reasonable justification, analysis and interpretation.
Satisfactory	3.5–5.4	Basic understanding; explanations are mostly descriptive and lack deeper reasoning.
Needs Improvement	0–3.4	Limited understanding; memorised answers; weak justification or interpretation; heavy dependence on team members.

Final Marks and Sign-off

Review 1 Marks (out of 7.5)	_____
Review 2 Marks (out of 7.5)	_____
Total (out of 15)	_____
Faculty Signature	_____
Student Signature	_____

Submission Checklist

☑ Problem statement and objectives are clearly stated.

☑ Dataset source, size, features and classes are documented.

☑ Metadata and preprocessing choices are explained and justified.

☑ Scope, impact, limitations and novelty are discussed.

☑ Model working principle and selection rationale are explained.

☑ Evaluation strategy, metrics, graphs and confusion matrix (if applicable) are included.

☑ Results are critically analysed and evidence-based inferences are provided.

☑ References are included and the student can explain their individual contribution.

Faculty evaluation should assess individual understanding, reasoning and interpretation. Technical execution or high accuracy alone should not determine the marks.